



14.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning objective.



I need help.



I got it.



I could help someone else.

Determine the value of an unknown decimal or fraction in an equation.	  
Translate a written description into an algebraic expression.	  
Write a description within a real-world context given an algebraic expression.	  
Write and solve one-step equations in one variable using addition and subtraction.	  
Write and solve one-step equations in one variable using multiplication and division.	  

Unit 14, Lesson 9: Solving One-Step Inequalities Using Addition and Subtraction



Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.



Learning Targets

- ☐ I can relate the solution set of an inequality to its graph on a number line.
- ☐ I can relate the solution set of an inequality to the context it represents.
- ☐ I can determine the minimum or maximum value that makes an inequality true.
- ☐ I can solve one-step inequalities in one variable using addition or subtraction.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.



14.9.1 Warm-Up: Which One Doesn't Belong?

Four inequalities are shown.

Inequality A	Inequality B	Inequality C	Inequality D
$x \geq 5$	$12 \leq 7$	$2 > x$	$x > 1$

1. Javier says that Inequality B does not belong with the others. What is one reason Javier might have said this?
2. Select a different inequality and explain why it doesn't belong.



14.9.2 Exploration: Representing Inequalities in Context

Height restrictions are often in place for roller coasters and rides at amusement parks to keep guests safe. At Disney's Typhoon Lagoon Water Park, the following height restrictions are in place.

Attraction	Height Restriction
Bay Slides	Guests must be 60 inches or shorter
Crush 'n' Gusher	Guests must be 48 inches or taller
Ketchakiddee Creek	Guests must be 48 inches or shorter

1. Work with your partner to determine which graph represents the height restrictions of each attraction at Typhoon Lagoon. Then, draw an arrow from each attraction to its matching graph.

Attraction	Graph Representing Safe Heights for the Attraction
Bay Slides	
Crush 'n' Gusher	
Ketchakiddee Creek	

2. Nick is happy to realize that he is tall enough to go on the Crush 'n' Gusher raft ride. Nick's friend Matt is 2 inches shorter than Nick. Explain whether Matt is tall enough to go on the ride with Nick.

3. Complete the statements.

If Nick is n inches tall, then Matt is ☐ $n + 2$
☐ $n - 2$ inches tall.

For Matt to be tall enough to go on Crush 'n' Gusher,

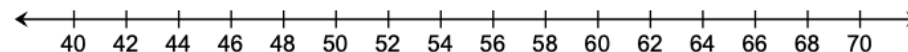
the value of ☐ $n + 2$
☐ $n - 2$ must be ☐ less than
☐ greater than or

equal to ☐ 48.
☐ 60. Therefore, the solutions to the

inequality ☐ $n + 2 \leq$
☐ $n + 2 \geq$
☐ $n - 2 \leq$
☐ $n - 2 \geq$ ☐ 48
☐ 60 represent Nick's height

that mean Matt could also ride the Crush 'n' Gusher.

4. On the number line, show all the possible heights that Nick could be if Matt is also tall enough to go on the ride.



5. What is the minimum height Nick must be in order for Matt to be able to go on the ride as well?

Julia noticed the graph also represents the solutions to the inequality $n \geq 50$.

6. Work with your partner to explain why the solutions to the inequality $n - 2 \geq 48$ can be represented the same way on a number line as $n \geq 50$.



14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Inequalities have an infinite number of solutions. The set of all the solutions of an inequality is called the **solution set**.

The **solution set** of an inequality includes the values that, when substituted for the variable, make the inequality true.

One way to represent the solution set of an inequality is by shading the values on the number line, beginning with the minimum or maximum value, and including all the values to the right or left of it that are included in the solution set.

1. Complete each statement.

A minimum or maximum value represented by an open circle means the value is ☐ included in ☐ excluded from the solution set.

A minimum or maximum value represented by a closed circle means the value is ☐ included in ☐ excluded from the solution set.

Graphs representing the solution set of inequalities are helpful when interpreting them in real-world contexts. However, sometimes there are parts of the solution set that may not make sense in a given context.

For example, to ride the Bay Slides at Typhoon Lagoon, guests must be 60 inches or shorter. The graph below represents the graph of $h \leq 60$ in this situation, where h is the height of guests that can ride the Bay Slides.



2. What does 60 represent in this situation?
3. What does the arrow pointing to the left on the number line represent?
4. Describe at least two values that are part of the solution set of $h \leq 60$ that do not make sense in this context.



Sometimes there are values that are part of the mathematical solution set that do not make sense in the context. In those cases, the variable may have a **constraint** (restriction or limitation) placed on it so that the mathematical model is realistic.

The Exploration demonstrated the inequalities $n - 2 \geq 48$ and $n \geq 50$ are equivalent because all of the same values of n make both inequalities true. Both inequalities were also represented using the same graph.

Inequalities that have the same solution set are called equivalent inequalities.



14.9.4 Exploration: Properties of Inequality

Three inequalities are shown.

$$x > -2.3 \qquad -2.3 > x \qquad -2.3 < x$$

1. Circle the inequalities that represent the same solution set.
2. Complete the statement.

For both inequalities, all the values of x that are _____ than _____ make both inequalities true.

These two inequalities are equivalent by the **asymmetric property of inequality**.



The **asymmetric property of inequality** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \leq \frac{7}{10}$.

4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides		
$-5 < \frac{3}{4}$	Subtract 2 from both sides		
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides		
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides		

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, ...
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, ...



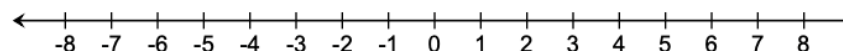
14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction

1. Complete the table to determine the value of $x - 3$ for each value of x .

x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x - 3$	-7		-5				-1		1

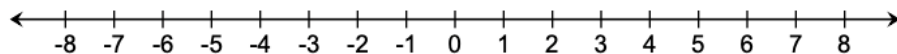
2. For which value of x is the equation $x - 3 = -2$ true?

3. Represent the solution to the equation $x - 3 = -2$ on the number line.



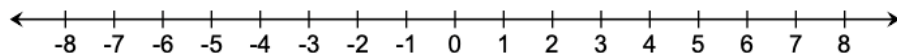
4. For which values of x in the table is the inequality $x - 3 > -2$ true?

5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.



The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.

6. Graph the solution set for the inequality.



7. Write a value on the line to show an equivalent inequality represented by the graph above.

$$x > \underline{\hspace{1cm}}$$

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x ?

Adding the same value to both sides of an inequality can be used to solve an inequality by the ***addition property of inequality***.



Addition Property of Inequality

If $a > b$, then $a + c > b + c$.

The ***addition property of inequality*** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The ***subtraction property of inequality*** can also be used to solve an inequality.

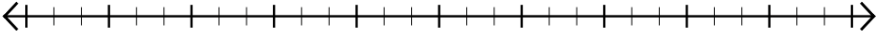


Subtraction Property of Inequality

If $a > b$, then $a - c > b - c$.

The ***subtraction property of inequality*** states that if the same value is subtracted from both sides of an inequality, the result will be an _____ inequality that is also _____.

These properties can be used to solve one-step inequalities algebraically.

9. Use the inequality $-12.25 > y - 2.4$ to complete the following.
- a. Find the solution set by isolating the variable. Show your work.
- b. Graph the solution set on the number line.
- 
- c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.



14.9.6 Wrap-Up: Cool Down

1. Complete the table.

The terms and information used in today's lesson that I knew were...	Something I wondered during today's lesson was...	Something I learned during today's lesson was...

